PATHSEEKER

App Proposal

TEAM MEMBER:

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Course and Section Number: 4495-002

Team Lead: Saruchi Sharma

Introduction

In today's competitive job market, people find it difficult to identify a career option that highly meets their skill set, interest, and market demand. The PathSeeker app will make personalized suggestions for users based on their profiles regarding educational background, skills, and preferences. The app will bridge the gap between seekers and available opportunities by offering data-driven recommendations.

Unemployment remains a critical challenge globally, with individuals often facing barriers to accessing suitable job opportunities. This project aims to address this gap by using artificial intelligence (AI) and user profiles to generate accurate career suggestions and job opportunities.

Problem Statement

Most candidates do not know how their unique skills and preferences relate to career possibilities. Traditional job boards cannot offer personalized career guidance; hence, users are overwhelmed by the choice uncertainty. The following are the research questions that this study tries to answer:

- How can we effectively map user profiles onto potential career paths?
- What features could be added to make career recommendations more accurate and relevant?
- How does PathSeeker contribute to long-term career planning?

Initial Hypotheses and Assumptions

1. Personalized career recommendations will increase user satisfaction and enhance decision- making.
2. Integration of real-time job market trends with user profiling will make the suggestions more relevant.
3. Upskilling resources will equip users to take recommended paths.

Potential Benefits

The benefits that PathSeeker would provide:

- It enables users to make well-informed choices in careers matching their profiles.
- It presents pathways for upskilling and career development clearly.
- It links users to resources and opportunities related to their career goals.

Proposed Research Project

Objectives:

- Develop a mobile application for career recommendations.
- Utilize AI to analyze user input and suggest relevant career paths.
- Integrate general question inputs for enhanced personalization.
- Provide insights into market demand and career growth potential.

Methodology:

- **Data Collection:**
  - User-provided data (skills, education, interests, ordinary questions).
  - Job market trends from online sources and databases.

Technology Stack:

- **Operating System:** Android
- **Programming Languages:** Python (for AI/ML), Dart (Flutter for frontend) or React Native
- **Frameworks:** Flutter (Mobile App)
- **Database:** Firebase for real-time storage, MongoDB for recommendations caching

Data Analysis Techniques:

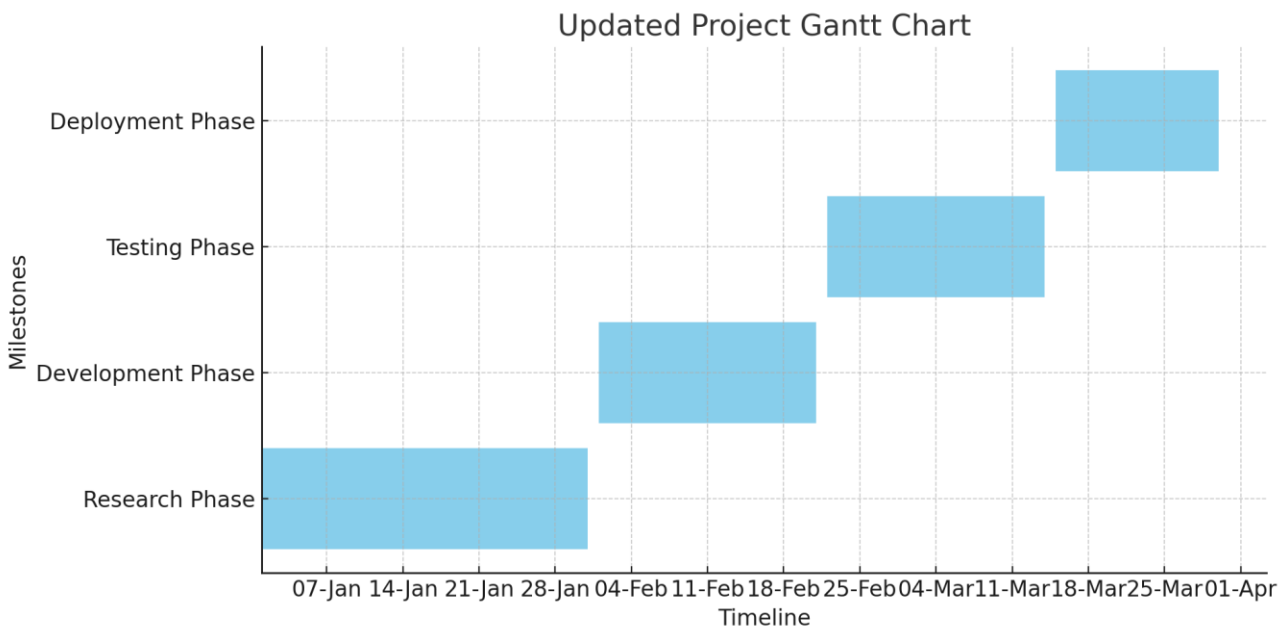
- Natural Language Processing (NLP)
- Machine Learning algorithms (Random Forest, K-Nearest Neighbors)
- Recommendation algorithms (Collaborative Filtering, Content-Based Filtering)

Expected Results:

- A user-friendly mobile career guide application.
- AI-powered career recommendations based on general question inputs.
- Increased user engagement and satisfaction in career decision-making.

Project Planning and Timeline

Milestone	Description	Deadline
Research Phase	Data collection and literature review	[31/01/25]
Development Phase	Backend & frontend development	[21/02/25]
Testing Phase	Usability and performance testing	[14/03/25]
Deployment Phase	Launch and feedback collection	[30/03/25]



Responsibilities

- **Saruchi:** Coordination, backend development, and overall project management.
- **Ajaypal:** UI/UX design and usability

testing. Both will work database integrations.

Work Date/Hours Logs

Date	Number of Hours	Description of Work Done
Jan 15, 2025	3	Spent time to think on the ideas we can work on
Jan 17, 2025	2	Discussed our final project ideas and required applications
Jan 18, 2025	2	Looked at other projects and basic functionalities
Jan 20, 2025	3	Tried to implement UI mockups

Project Contract

We, the undersigned, agree to adhere to the scope, timelines, and deliverables outlined in this proposal. We commit to regular team meetings and transparent communication to ensure project success.

Each team member agrees to contribute a minimum of 120 hours individually towards the completion of the project, distributed across the project phases as required. All members are responsible for managing their time effectively and ensuring that their contributions align with the project milestones.

Team:

Saruchi Sharma

Ajaypal Singh

Acknowledgement

We thank Padam Priya, our professor, for continuous guidance and feedback for the project ideas and motivating us to work on this project.